

Effect Heterogeneity of the Obesity Paradox in ICU Mortality

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Abstract

This study examines whether the effect of obesity on in-hospital mortality varies across ICU patient subgroups using the MIMIC-IV database. Under a causal framework, we estimate subgroup-specific effects via parametric g-computation after adjusting for baseline confounders. We find that obesity is associated with lower mortality overall, consistent with the obesity paradox, but the effect varies across subgroups. The protective association is stronger among older patients and those with moderate illness severity, while differences by sex are minimal. These findings highlight that the relationship between obesity and mortality is heterogeneous across ICU populations.

1 Introduction

The association between obesity and mortality among critically ill patients has been extensively investigated, with obesity commonly recognized as a risk factor for numerous adverse health outcomes. Previous studies have however identified a phenomenon known as the obesity paradox, in which obese individuals appear to exhibit lower mortality risk in specific clinical contexts. In this study, we will examine the obesity paradox and whether the causal effect of obesity on intensive care unit (ICU) mortality varies across different subgroups.

2 Data Description

This study uses data from the Medical Information Mart for Intensive Care IV (MIMIC-IV), a publicly available database of de-identified electronic health records from intensive care unit (ICU) patients at Beth Israel Deaconess Medical Center.

We focus on ICU patients aged 18 years or older and construct a dataset to examine the relationship between obesity and mortality. The primary outcome is in-hospital mortality,

defined as a binary indicator of mortality during the ICU stay. We focus on this outcome because it is fully observed and directly reflects outcomes during illness, whereas post-discharge mortality may be subject to incomplete follow-up and additional confounding from factors not captured in the data. The treatment variable is obesity status, defined using body mass index (BMI): obese ($\text{BMI} \geq 30$) versus non-obese ($\text{BMI} < 30$).

To address confounding, we include covariates that may influence both obesity and mortality: age (three categories: < 40 , $40\text{--}64$, ≥ 65 years), sex (binary), illness severity (Sequential Organ Failure Assessment [SOFA] score), and ICU type (ward classification). These variables are selected based on clinical relevance and their roles in the causal structure. Age and sex are confounders affecting both obesity and mortality risk. Illness severity and ICU type directly influence mortality and may also be associated with obesity, as obesity-related comorbidities can affect admission severity and ICU placement, while ICU type captures differences in clinical management and surgical case mix relevant to both.

3 Causal Model and Identification

Under the assumed causal framework, the effect of obesity on ICU mortality is potentially confounded by demographic and clinical variables. We assume that the causal effect is identifiable by adjusting for age, sex, severity at ICU admission, and ICU type, which satisfy the back-door criterion. The causal query of interest is the subgroup-specific effect of obesity on in-hospital mortality, defined as

$$E(Y_1 \mid V = v) - E(Y_0 \mid V = v),$$

which represents the difference in mortality if all individuals in subgroup v were obese compared to if none were obese.

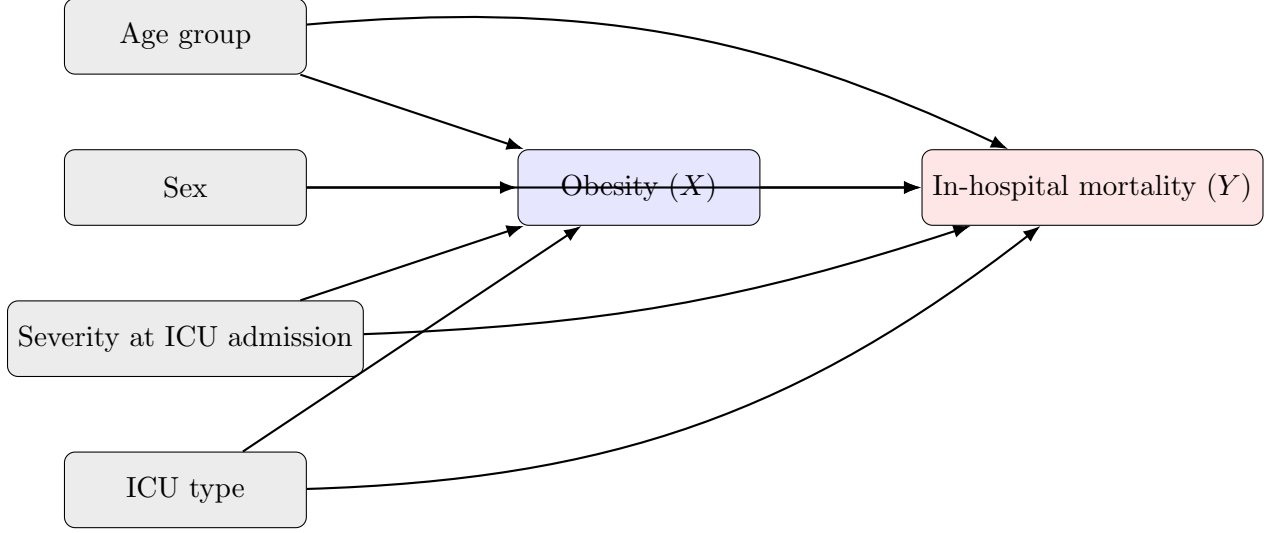


Figure 1: DAG

The DAG in Figure 1 represents the assumed causal model for the effect of obesity (X) on in-hospital mortality (Y).

The primary causal effect of interest is the effect of obesity on mortality, represented by the arrow $X \rightarrow Y$. However, several baseline variables may act as common causes of both the exposure and the outcome, thereby inducing confounding. The four variables are assumed to influence both obesity status and mortality, therefore opening backdoor paths between X and Y (e.g., $X \leftarrow \text{age} \rightarrow Y$), which can bias the estimation of the causal effect if not properly controlled.

4 Identification, Estimation, and Inference

4.1 Identification

Based on the DAG in Figure 1, age group, sex, severity at ICU admission, and ICU type are treated as baseline confounders. Adjusting for these variables blocks all backdoor paths between obesity and mortality. We assume conditional exchangeability ($Y_x \perp X \mid C$), consistency, and positivity, under which the causal effect is identifiable.

We focus on effect heterogeneity using the subgroup-specific risk difference:

$$\tau(v) = E(Y_1 \mid V = v) - E(Y_0 \mid V = v).$$

4.2 Estimation

We estimate $\tau(v)$ using parametric g-computation via standardization. For each effect modifier, we fit a separate logistic regression model including obesity, the modifier, their interaction, and baseline confounders.

For each subgroup level v , we set obesity to $X = 1$ and $X = 0$ for all individuals in that subgroup, obtain predicted mortality probabilities from the fitted model, and average these predictions. The difference between these averages yields $\hat{\tau}(v)$ on the risk difference scale.

4.3 Inference and Sensitivity

We quantify uncertainty using a nonparametric bootstrap with $B = 200$ resamples. In each resample, we repeat the full estimation procedure and use the resulting distribution of estimates to construct 95% confidence intervals. Although we adjust for observed confounders, our results may still be affected by unmeasured factors. In particular, severity at ICU admission may not be strictly pre-treatment and could itself be influenced by obesity, so adjusting for it may introduce bias.

5 Data and Preprocessing

The data were obtained from the MIMIC-IV database and processed to create an analysis-ready dataset at the ICU stay level. Observations with missing or invalid values in key variables were removed. Height and weight measurements were used to compute BMI for obesity classification. Age was categorized as under 40, 40–64, and 65 years or older. Illness severity, based on SOFA score, was stratified into low, moderate, and high categories. Infrequent ICU types were consolidated into an “Other” category, with the remaining categories including cardiac, medical (MICU), neuro, and surgical (SICU).

6 Results

Table 1: Baseline characteristics by obesity status

Variable	Category	Non-obese	Obese
Sample size		32070	17250
Mortality rate		0.150	0.141
Age group	< 40	0.081	0.073
	40–64	0.337	0.437
	65+	0.583	0.489
Sex	F	0.385	0.420
	M	0.615	0.580
Severity	Low	0.543	0.504
	Medium	0.337	0.340
	High	0.120	0.156
ICU type	Cardiac	0.399	0.416
	MICU	0.325	0.331
	Neuro	0.042	0.033
	Other	0.004	0.007
	SICU	0.231	0.214

Table 2: Subgroup-specific causal effects of obesity on in-hospital mortality

Modifier	Level	$E(Y X = 1)$	$E(Y X = 0)$	Risk Diff	95% CI
Age group	< 40	0.087	0.090	-0.003	[-0.021, 0.016]
	40–64	0.112	0.127	-0.016	[-0.023, -0.006]
	65+	0.157	0.180	-0.023	[-0.030, -0.014]
Sex	F	0.149	0.167	-0.018	[-0.026, -0.008]
	M	0.125	0.144	-0.019	[-0.027, -0.011]
Severity	Low	0.048	0.063	-0.015	[-0.022, -0.009]
	Medium	0.143	0.168	-0.025	[-0.036, -0.012]
	High	0.458	0.474	-0.017	[-0.038, 0.008]
ICU type	Cardiac	0.077	0.082	-0.005	[-0.010, 0.002]
	MICU	0.202	0.234	-0.033	[-0.044, -0.023]
	Neuro	0.035	0.042	-0.007	[-0.022, 0.011]
	SICU	0.157	0.180	-0.023	[-0.038, -0.009]
	Other	0.251	0.370	-0.118	[-0.232, -0.022]

Table 1 presents baseline characteristics by obesity status while Table 2 presents adjusted subgroup-specific causal effects obtained after controlling for these baseline variables. Comparing the two tables highlights the importance of adjustment: while Table 1 shows raw

differences in patient characteristics, Table 2 isolates the causal effect of obesity on mortality within subgroups.

Across most subgroups, the estimated risk differences are negative, indicating that obesity is associated with a lower risk of mortality after adjustment for baseline covariates. There is also clear evidence of effect heterogeneity across subgroups. The paradoxical effect is stronger in older patients, with the largest reduction in mortality observed among those aged 65 and above, while the effect is small and not statistically significant among patients under 40. In contrast, the effect is similar for males and females, suggesting little heterogeneity by sex.

For illness severity, the effect appears strongest in patients with medium severity and weaker in those with low or high severity, with the high severity group showing a confidence interval that includes zero. This suggests that the protective association of obesity may vary with baseline illness severity.

Substantial heterogeneity is also observed across ICU types. The effect is strongest in the medical ICU (MICU) and surgical ICU (SICU), while it is small and not statistically significant in cardiac and neuro ICUs. The estimate for the “Other” category is large but has a wide confidence interval, indicating instability due to a small sample size.

Overall, the results suggest that obesity is associated with a reduction in in-hospital mortality across the studied sub-groups. However, the magnitude of this effect varies by age, severity, and ICU type, highlighting the importance of accounting for effect heterogeneity in causal analyses.

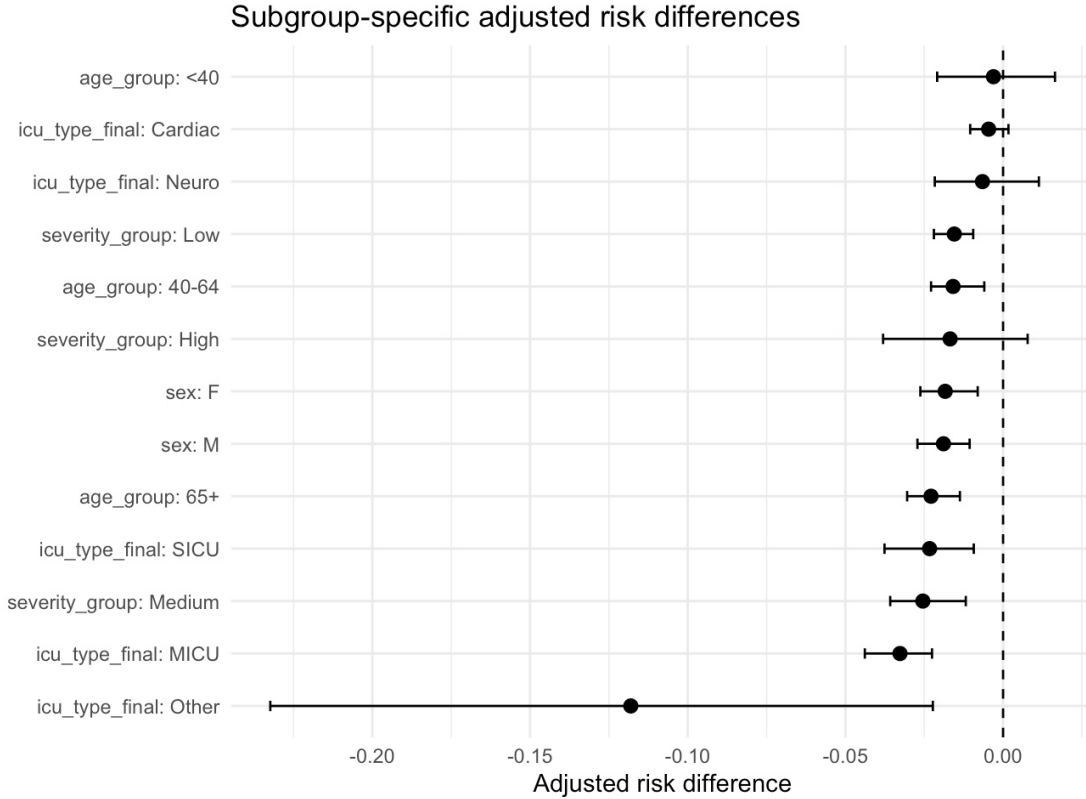


Figure 2: Subgroup-specific causal effects of obesity on in-hospital mortality.

Figure 2 provides a visual summary of the subgroup-specific effects and reveals several important patterns that are less apparent from the table alone. First, most point estimates lie to the left of zero, reinforcing the overall protective association of obesity with mortality. However, the magnitude of this effect varies substantially across subgroups, indicating meaningful effect heterogeneity.

In particular, the protective effect appears stronger in subgroups with higher baseline risk, such as older patients and those in the MICU and SICU. This is consistent with the idea that the obesity paradox may be more obvious in populations with greater underlying vulnerability, where additional physiological reserves could play a larger role. In contrast, subgroups such as younger patients, cardiac ICU, and neuro ICU show estimates closer to zero with confidence intervals overlapping zero, suggesting little to no clear effect.

The forest plot also highlights the role of uncertainty. Some subgroups, especially the “Other” ICU category, exhibit wide confidence intervals, indicating instability likely due to smaller sample sizes. This suggests that extreme estimates in these groups should be interpreted cautiously.

The visualization supports the interpretation that the obesity paradox is not uniform

across all patients, but instead is concentrated in specific clinical contexts, particularly among older and moderately to severely ill patients.

7 Conclusion

This study estimates subgroup-specific effects of obesity on ICU mortality and finds evidence consistent with the obesity paradox.

However, this effect is not uniform across patients. The protective association is stronger among older individuals and those with moderate illness severity, and varies across ICU types, while differences by sex are minimal. This is consistent with prior work showing improved survival among obese patients in older populations, suggesting that the obesity paradox may be more pronounced in higher-risk groups (Tian et al., 2024).

Overall, our results suggest that the relationship between obesity and ICU mortality depends on patient characteristics and clinical context. While obesity appears protective in some groups, these findings should be interpreted cautiously given the observational nature of the study and the potential for residual confounding.

8 Discussion and Limitations

One possible explanation for the paradox is that excess body mass provides metabolic reserve during critical illness, allowing patients to better tolerate the acute catabolic state (Wang et al., 2025).

Despite these findings, several limitations should be noted. First, the analysis relies on the assumption of no unmeasured confounding, which may be violated due to missing factors such as comorbidities, functional status (measured by Activities of Daily Living), or differences in treatment decisions across patients. Second, severity at ICU admission may not be strictly pre-treatment, as it could be influenced by obesity itself or by differences in admission timing and care patterns; prior work has shown that obese patients tend to have greater ICU utilization and different treatment strategies, which may affect observed outcomes (Lim et al., 2022). Conditioning on such a variable may therefore introduce bias. Third, some subgroup estimates are imprecise, particularly for smaller groups with wide confidence intervals, which limits the reliability of conclusions in those subpopulations.

References

Lim, A. K. H., Rathnasekara, G. K., Kanumuri, P., Malawaraarachchi, J. K., Song, Z., & Curtis, C. A. (2022). Goals of care, critical care utilization and clinical outcomes in obese patients admitted under general medicine. *Journal of Clinical Medicine*, 11(24), 7267.

Tian, J., Jin, K., Qian, H., & Xu, H. (2024). Impact of the obesity paradox on 28-day mortality in elderly patients critically ill with cardiogenic shock. *Diabetology & Metabolic Syndrome*.

Wang, Y., Li, D., & Zhu, L. (2025). Obesity paradox in critically ill patients: Do patients with critical diseases benefit from obesity? *Obesity Facts*. <https://doi.org/10.1159/000547813>

A Appendix

A.1 Estimation Procedure

We estimate subgroup-specific causal effects using parametric g-computation. For each effect modifier, we fit a logistic regression model including obesity, the subgroup variable, their interaction, and baseline confounders (age, sex, severity, and ICU type).

For each subgroup, we set obesity to $X = 1$ and $X = 0$, predict mortality probabilities using the fitted model, and average these predictions to obtain $\hat{E}(Y_1 | V = v)$ and $\hat{E}(Y_0 | V = v)$. The difference yields the estimated risk difference.

A.2 Inference

Uncertainty is quantified using a nonparametric bootstrap with $B = 200$ resamples. Confidence intervals are constructed using empirical percentiles.

A.3 Implementation

All analyses were conducted in R using the `glm()` function with a binomial family. Predictions were obtained using `predict(..., type = "response")`.